

CHEST PAIN

Clinical Question

Is the chest pain due to obstructive CAD?

CAD Consortium Clinical Score

Pre-test Probability of Obstructive (>50%) CAD:

Age, Sex
Chest Pain Character
Diabetes
Hypertension
Dyslipidemia
Smoking History

[Link to Calculator](#)

Will Aggressive Measures Reduce Risk of Short-Term MACE?

HEART Score

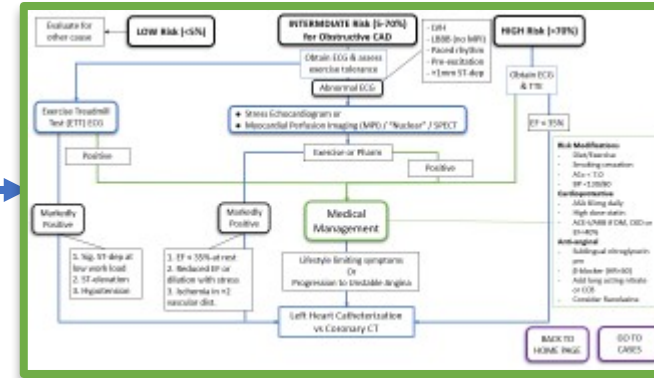
Six-week risk of MACE

History
ECG
Age
Risk Factors
Troponin

[Link to Calculator](#)

Diagnosis and Management

Chronic Coronary Artery Disease



Acute Coronary Syndrome (ACS)

Coming soon, as a separate talk...

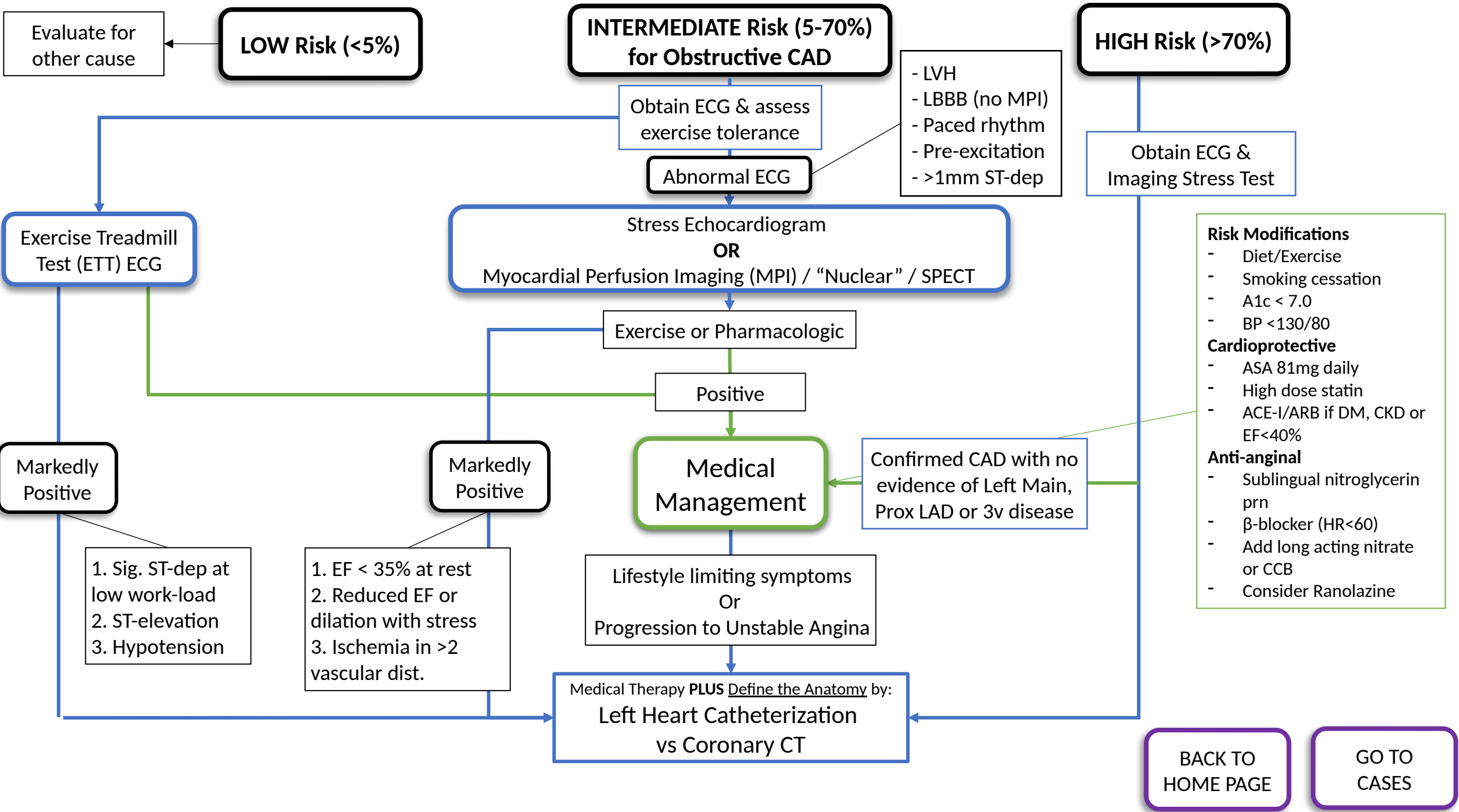
Goals of Therapy

Prevent & Treat

MACE

Heart Failure

Angina



Click

Case 1

73yo man, never smoker with HTN presents to clinic with one week of an ache and pressure just inferior to his sternum that comes on with minimal activity, usually in the evening or early in the morning. Self resolves after about 10 minutes. Still able to walk a few blocks with his dog.

Total Chol 180, HDL 38, LDL 142
A1c 5.6

Intermediate (33%)

Normal Sinus Rhythm

ECG Exercise Treadmill Test

Nml baseline ECG, resting BP 130/80, achieved 10 METS and 80% max HR, experiences chest tightness and test discontinued due to hypotension 90/50

Left Heart Cath

Case 2

45yo woman, remote smoker with well controlled DM presents to clinic with two months of sharp intermittent central chest pain every couple of days while working around the house or in the grocery store and goes away on its own with rest. She cannot climb stairs or walk more than a block due to her bilateral knee OA.

Total Chol 180, HDL 38, LDL 142
A1c 7.2

Low (5%)

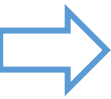
Normal Sinus Rhythm

No stress test vs.
pharm (dipyridamole) MPI

Normal resting MPI, indeterminant septal changes during stress with out cp, sob, or hemodynamic changes.

No additional treatment

Case 3



75yo man, active smoker with HTN and COPD (GOLD 3), presents to clinic with one week of chest tightness when gardening or bringing in the groceries that goes away after a few minutes of rest.

Total Chol 180, HDL 38, LDL 142 – on statin
A1c 5.6

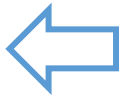
High (79%)

LVH, 1+mm ST dep in V4-6

Stress with imaging

Concentric LVH and grade I diastolic dysfunction. Mild apical hypokinesis, LV EF 55% at rest. Small area of inducible lateral wall ischemia and EF of 55% during stress.

Empiric medical mgmt for stable CAD



Case 4

62yo man, active smoker with COPD (GOLD 3), DM and a prior DES placed in his RCA three years ago now complaining of shortness of breath and chest tightness that occurs every couple of days, usually with exertion but sometimes with rest and goes away on its own. He cannot walk up stairs due to hip pain. On ASA, BB, ACE-I, statin. VSS, exp wheezing on exam.

On a statin & A1c – 7.9

??? - At least intermediate (42%)

Sinus with Q in II, III and aVF, incomplete RBBB

Pharm MPI vs. coronary CT vs. LHC

MPI – inducible large area anterior LV wall ischemia and reduction in EF from 55%→45% during stress

Likely clopidogrel, nitrate and LHC to r/o LM or prox LAD

Case 5

55yo woman, active smoker with well controlled DM presents to clinic with two to three weeks of chest tightness and shortness of breath that comes on when she is hiking and goes away with rest after 10-15 minutes.

Total Chol 180, HDL 38, LDL 142
A1c 7.2

Intermediate (22%)

Normal sinus rhythm

ECG Treadmill Exercise Test

Nml baseline ECG, resting BP 130/80, achieved 12 METS and achieved 80% max heart rate. Test discontinued due to chest tightness

Stress with imaging to r/o high grade lesion

Case 6



65yo man, active smoker with HTN presents with two days of chest pain whenever he goes up the stairs or after a few blocks of walking the dog. He feels diaphoretic and dizzy when it happens and needs to sit down. Symptoms self resolve after 10 minutes.

Total Chol 180, HDL 38, LDL 142 – on statin
A1c 5.6

High – Concerning for ACS

LVH, 1+mm ST dep and TWI V1-3, Trop neg

Admitted for ACS rule out with serial ECGs & trops on heparin gtt, inpatient TTE

Subsequent ECGs & trops without concerning changes. No recurrence of chest pain. TTE with mild LV hypertrophy, normal LV EF and normal valves.

Outpatient LHC

Guideline Update: 2021 Chest Pain & 2023 Chronic Coronary Disease

2021 AHA/ACC Chest Pain Guideline (Gulati et al.)

Describe chest pain as cardiac / possibly cardiac / noncardiac — retire "typical / atypical."

Stable pain, no known CAD, low pretest probability: defer testing; coronary artery calcium (CAC) score is a useful first-line gatekeeper to exclude calcific plaque.

Intermediate–high risk: CCTA is now Class 1 first-line (anatomic) alongside stress imaging (functional). CCTA favored if <65 y or less obstructive CAD suspected; stress favored if >65 y or more obstructive CAD suspected.

2023 AHA/ACC Chronic Coronary Disease Guideline (Virani et al.)

Beta-blockers not required as anti-anginal without prior MI or LVEF $\leq 50\%$; high-intensity statin first-line; add SGLT2i or GLP-1 RA in CCD + diabetes (SGLT2i if HF); colchicine 0.5 mg daily is reasonable (Class 2a).

ISCHEMIA trial (2020): in stable CAD with moderate–severe ischemia, an initial invasive strategy did not reduce death/MI vs optimal medical therapy — supports treating medically first and deferring cath.

Take-Home Points

1. Frame the question first.

Outpatient non-acute chest pain asks “is this obstructive CAD?” — distinct from the ED question “is this ACS?”

2. Risk-stratify by pretest probability (CAD Consortium Score).

Low <5%: look for an alternative cause. Intermediate 5–70%: stress test. High >70%: ECG + imaging to exclude left main / proximal LAD / 3-vessel disease, then empiric medical therapy.

3. Choose the stress test by what the patient can do.

Exercise treadmill ECG if the patient can exercise and the baseline ECG is interpretable; otherwise imaging stress (echo or MPI), exercise or pharmacologic.

4. Treat medically first; image the anatomy only when warranted.

Start guideline-directed medical therapy; reserve CCTA/left heart cath for markedly positive testing or failure of medical management (ISCHEMIA).

5. Scores support — never supersede — clinical judgment.

Attribution & Optimization Changelog

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Optimization (2026-06-10)

Corrected deck typo “INTERMDIATE” → “INTERMEDIATE” (slide 2). Appended this update + take-home slide; all original animated buttons/builds preserved.

Currency: 2019 ESC framework verified current; layered 2021 AHA/ACC Chest Pain & 2023 AHA/ACC Chronic Coronary Disease updates (all source-cited). No invented values.

Companion: Chest-Pain-CAD-eval-Learner-Handout-1.pdf — same INTERMDIATE→INTERMEDIATE typo corrected in place (filename kept); no other errors.